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Test Scope

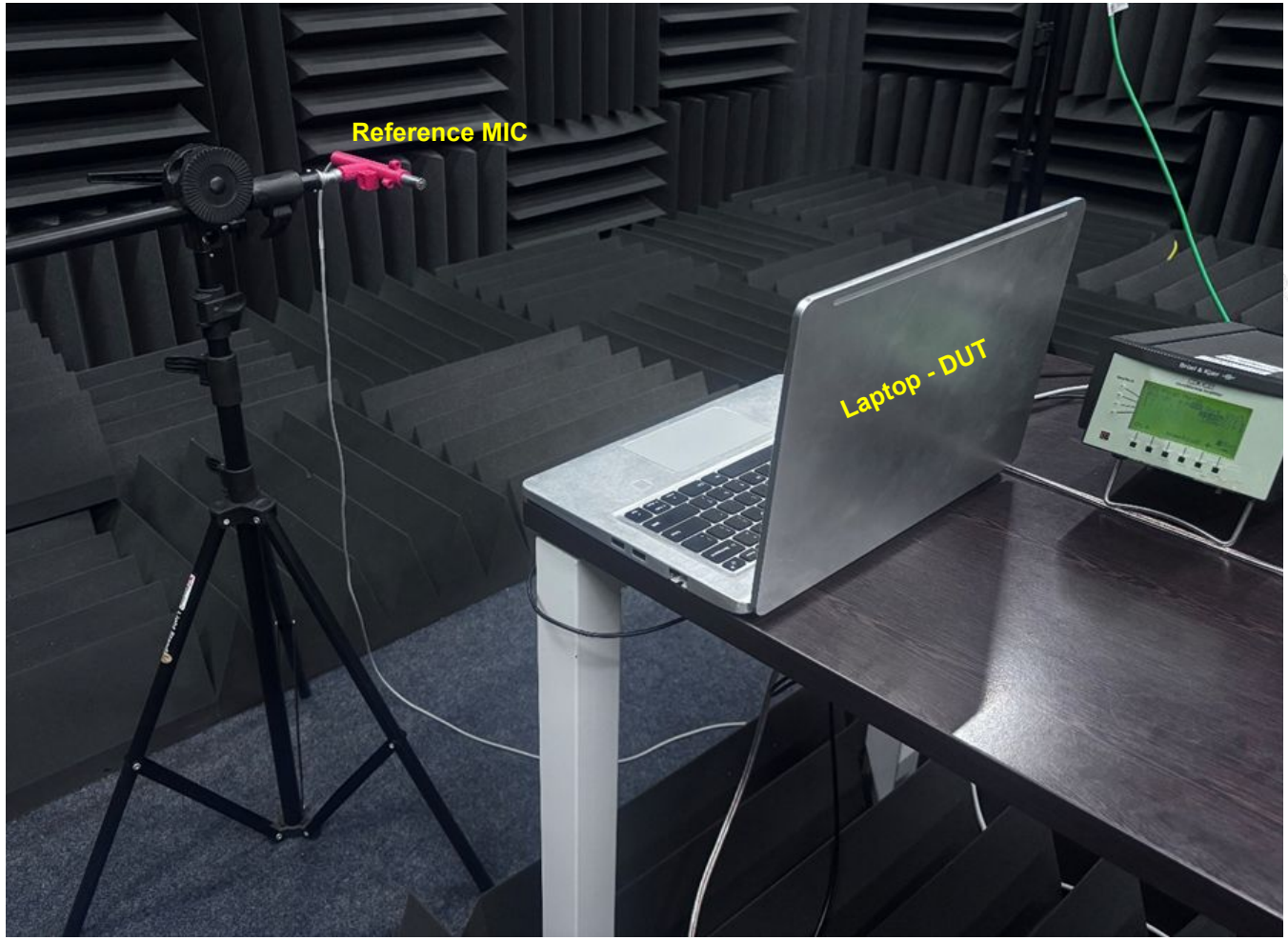
This report presents the validation test results of the laptop speaker under different test orientations and operating conditions. The measurements were performed using **SoundCheck** along with a calibrated measurement microphone to ensure accurate acoustic analysis.

The objective of this testing was to evaluate the speaker performance under multiple conditions and orientations. Tests were conducted at a distance of **0.5 m** between the Laptop(DUT) and the measurement microphone.

Issue Encountered and Resolution

- During the initial testing, it was observed that the **polarity of the right channel speaker was reversed**, which affected the acoustic output characteristics. This wiring discrepancy was identified during the validation process.
- After identifying the issue, the speaker wiring was corrected and the tests were re-conducted under the same test conditions to ensure accurate measurement results. The updated results obtained after the correction are presented in the following sections of this report.
- To avoid similar issues in future builds, the necessary **modifications should be implemented in the schematic design and/or wiring documentation** to ensure correct channel polarity during assembly.

Test Condition - 0.5m distance of Normal laptop working condition.



Test Type: Rear Radiation / Off-Axis Acoustic Measurement

Distance: 0.5 m between DUT and microphone

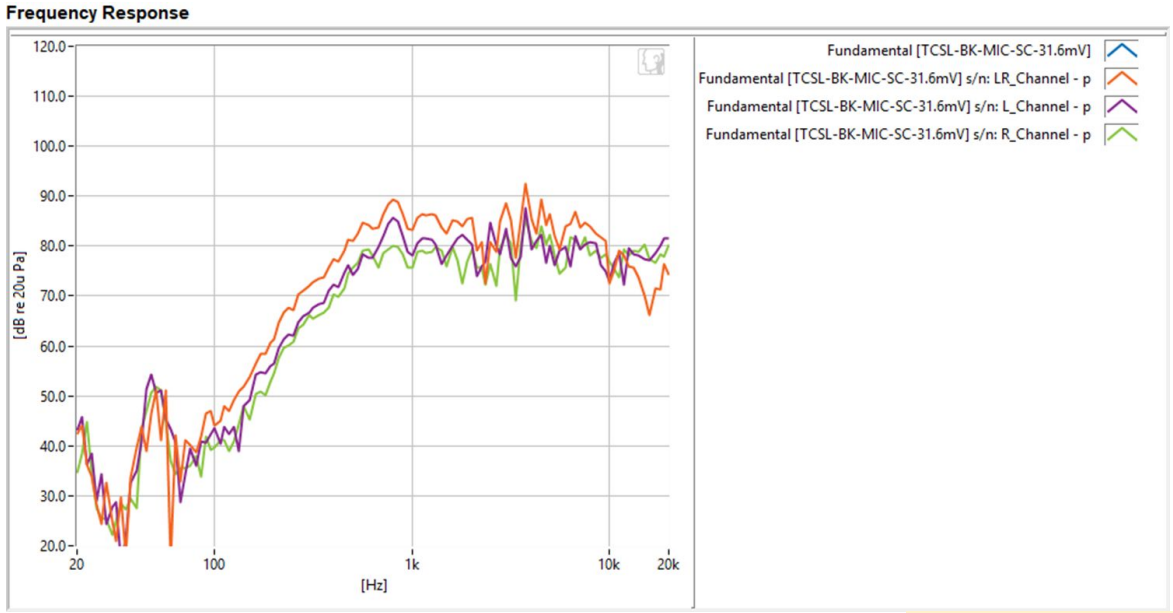
Environment: Acoustic chamber with sound-absorbing wedges

Orientation: The Rear side of the laptop chassis is facing the microphone.

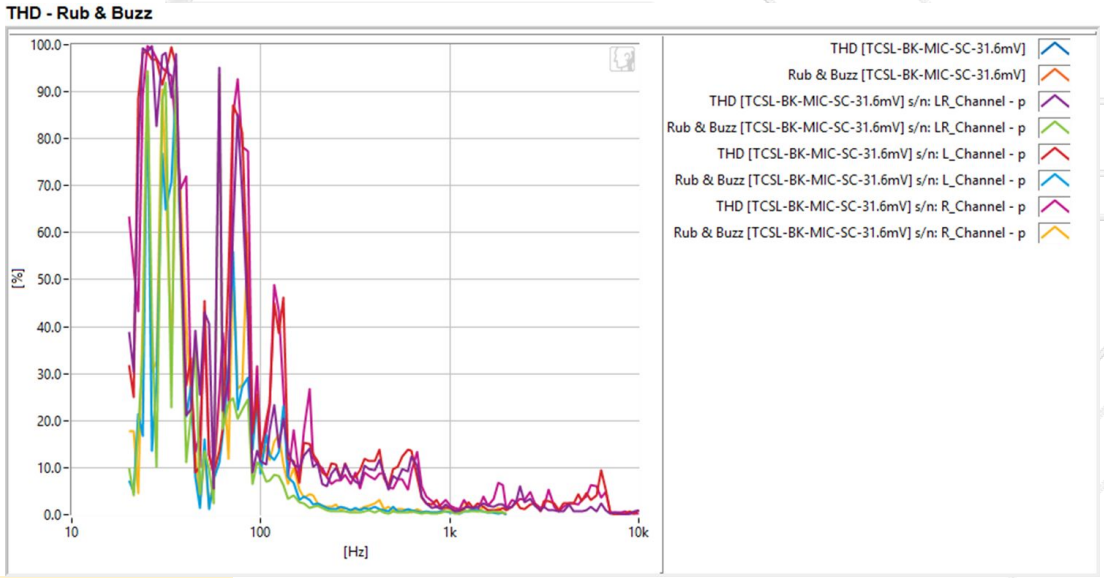
Laptop Condition: Internal audio playback enabled

Acoustic Measurement Results

Frequency Response



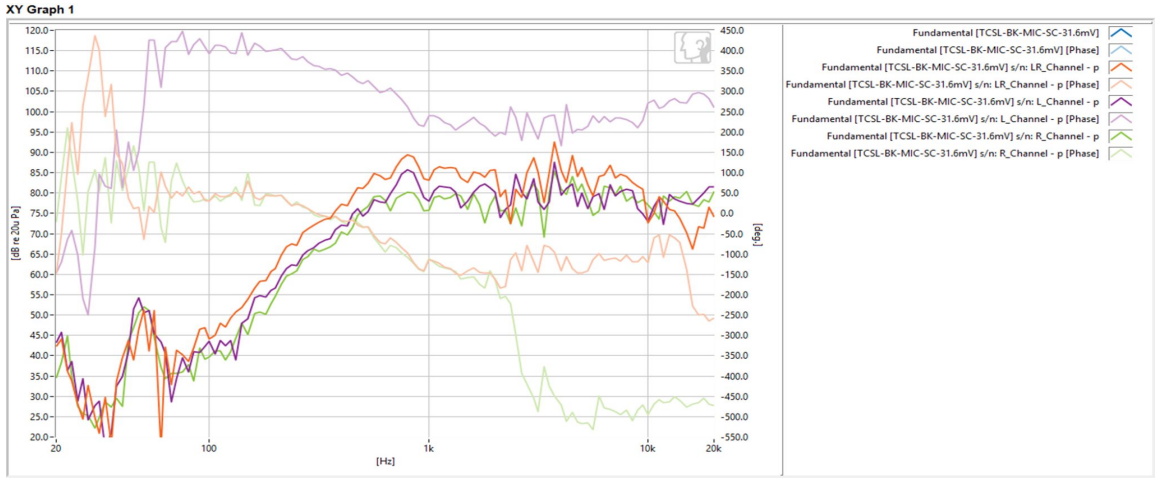
THD - Rub & Buzz



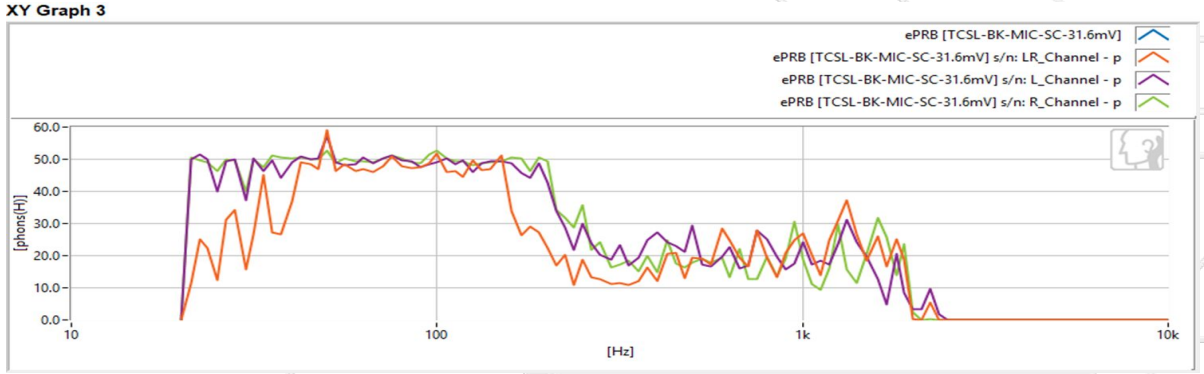
Average Loudness dB SPL

S.NO	Measurement Parameters	Value (dB re 20 uPa)
1	Average Loudness	76.5
2	Average Loudness s/n: Left_Channel	73.1
3	Average Loudness s/n: Right_Channel	71.7

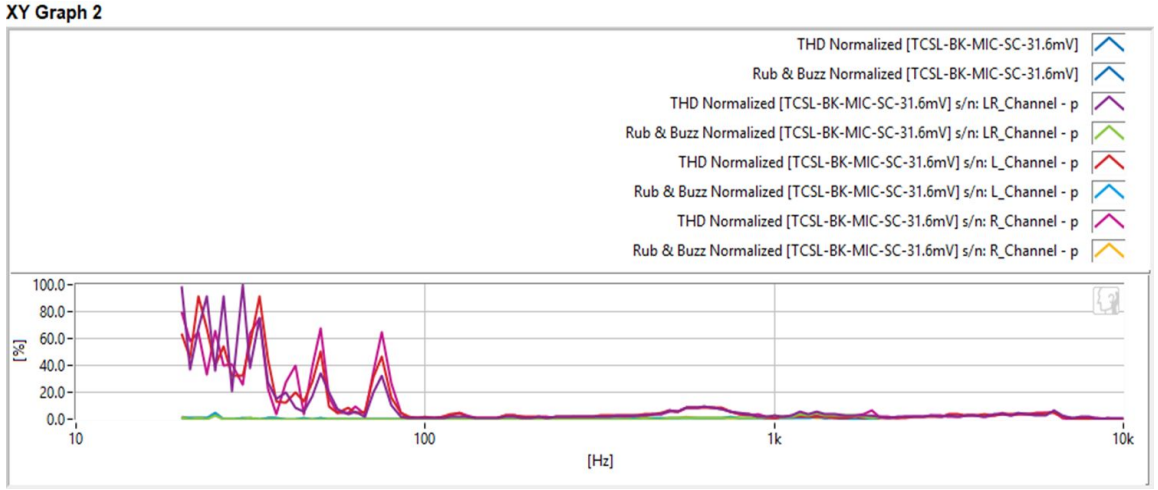
Phase Response



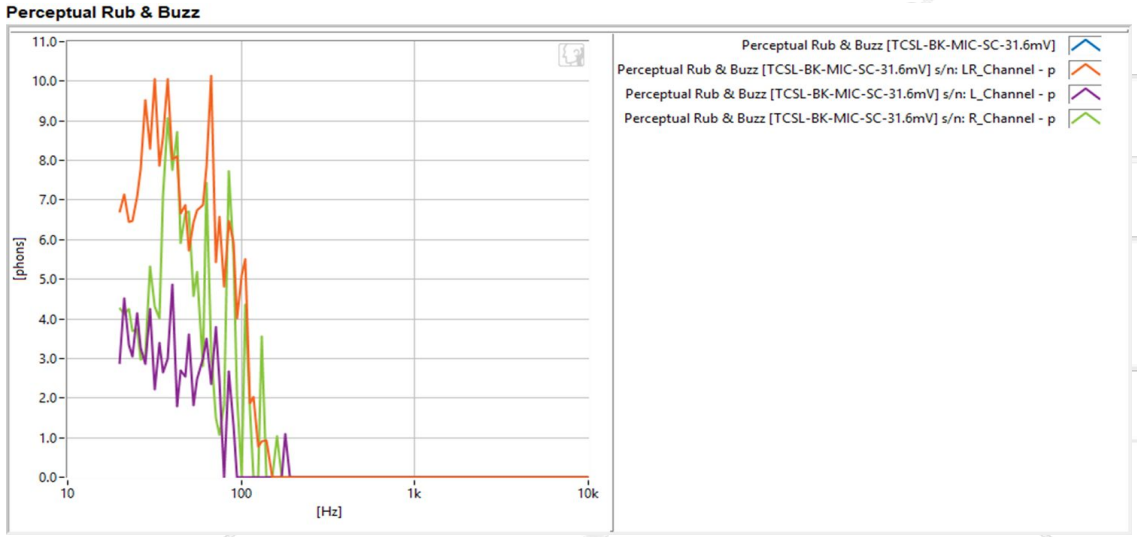
ePRB



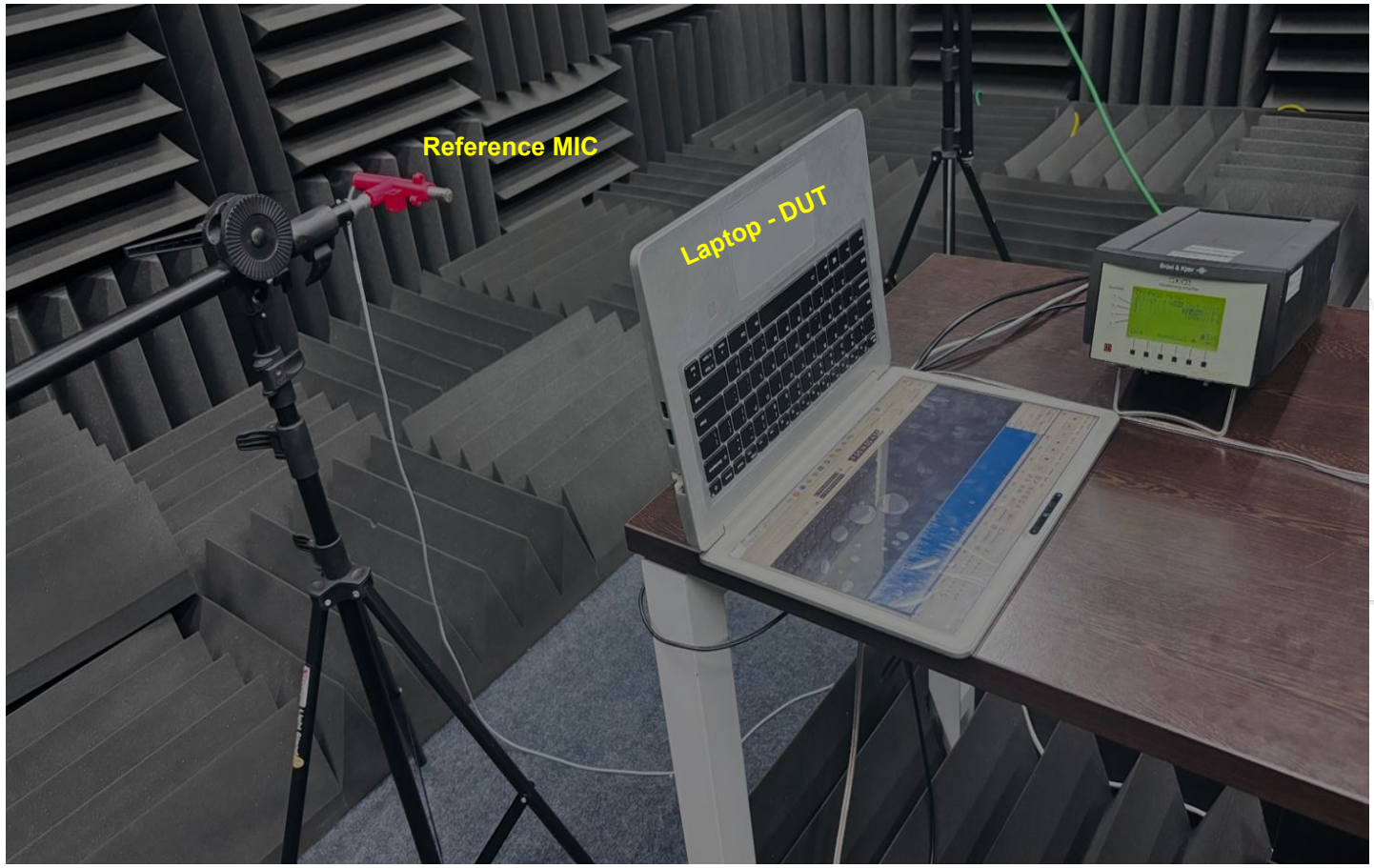
THD Normalized and Rub & Buzz Normalized



Perceptual Rub & Buzz



Test Condition - 0.5m distance, On-Axis Acoustic Measurement



Test Type: On-Axis Acoustic Measurement

Distance: 0.5 m between DUT and microphone

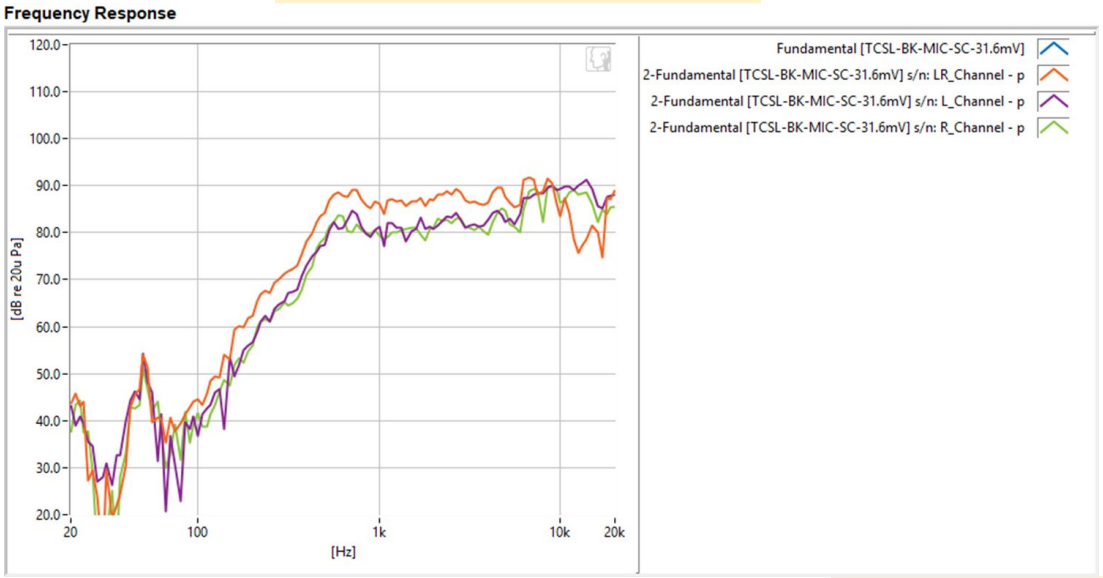
Environment: Acoustic chamber with sound-absorbing wedges

Orientation: The Front side of the laptop chassis is facing the microphone.

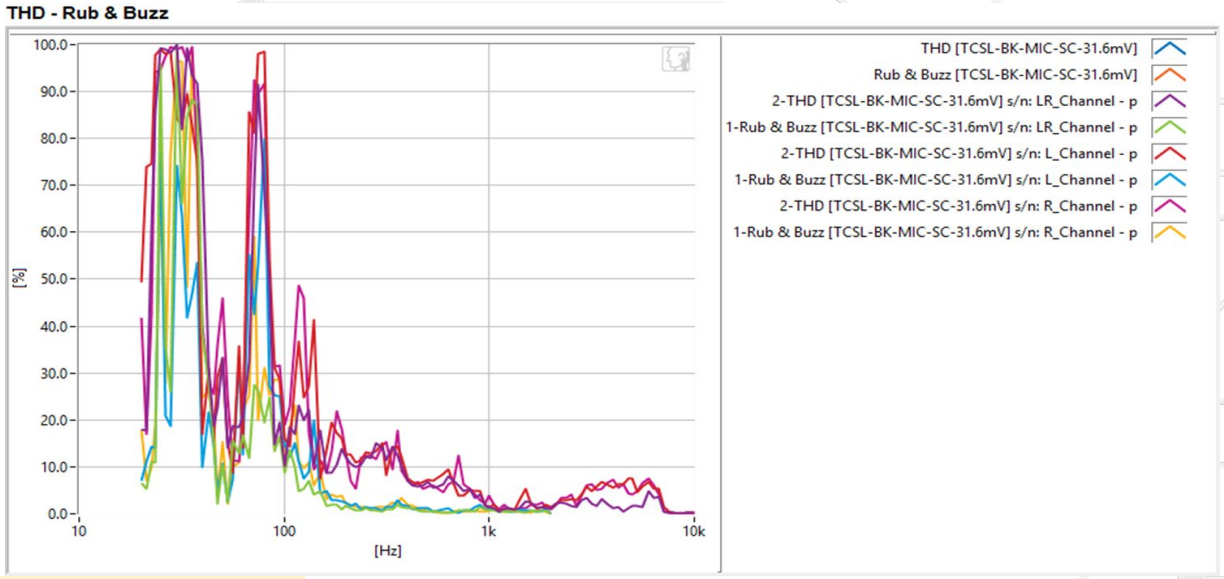
Laptop Condition: Internal audio playback enabled

Acoustic Measurement Results

Frequency Response



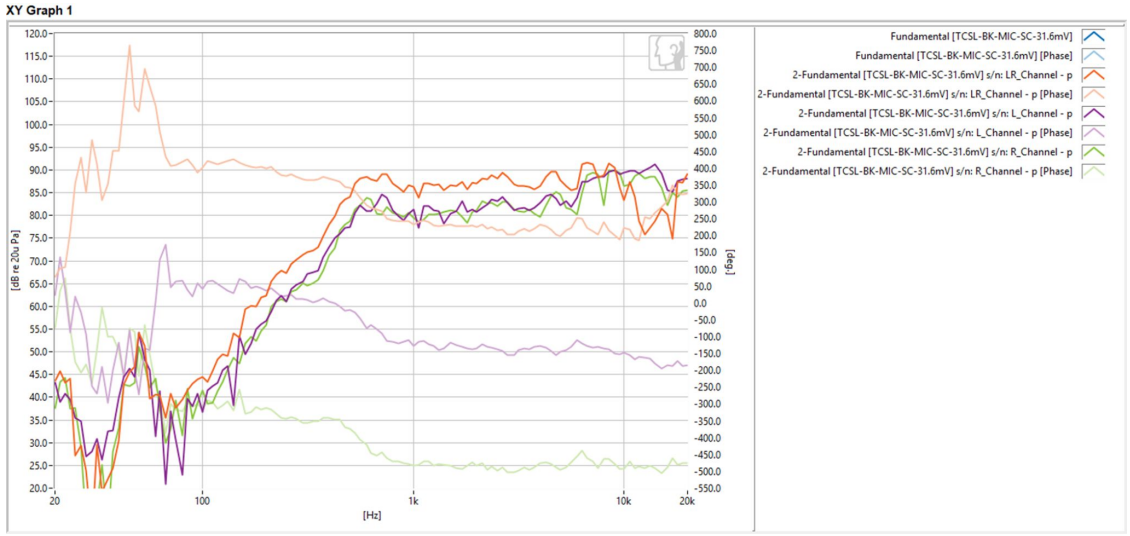
THD - Rub & Buzz



Average Loudness dB SPL

S.NO	Measurement Parameters	Value (dB re 20 uPa)
1	Average Loudness	79.6
2	Average Loudness s/n: Left_Channel	76.2
3	Average Loudness s/n: Right_Channel	75.4

Phase Response



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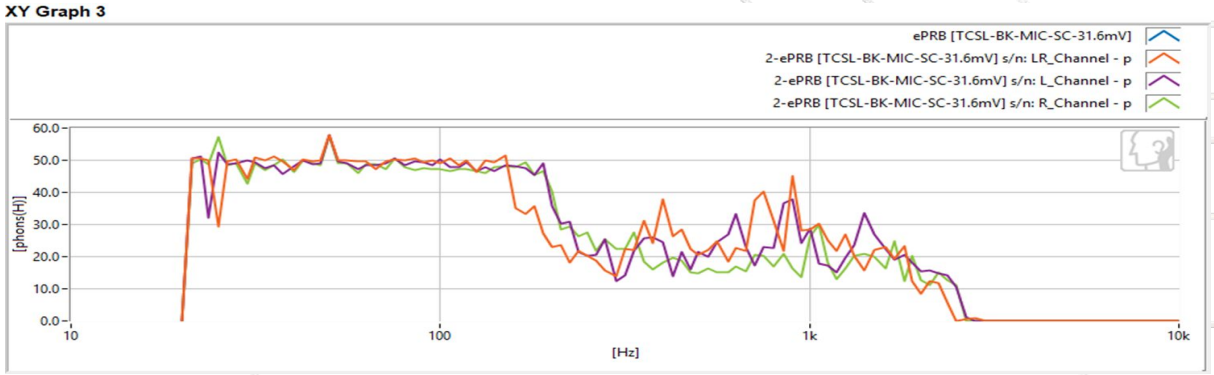
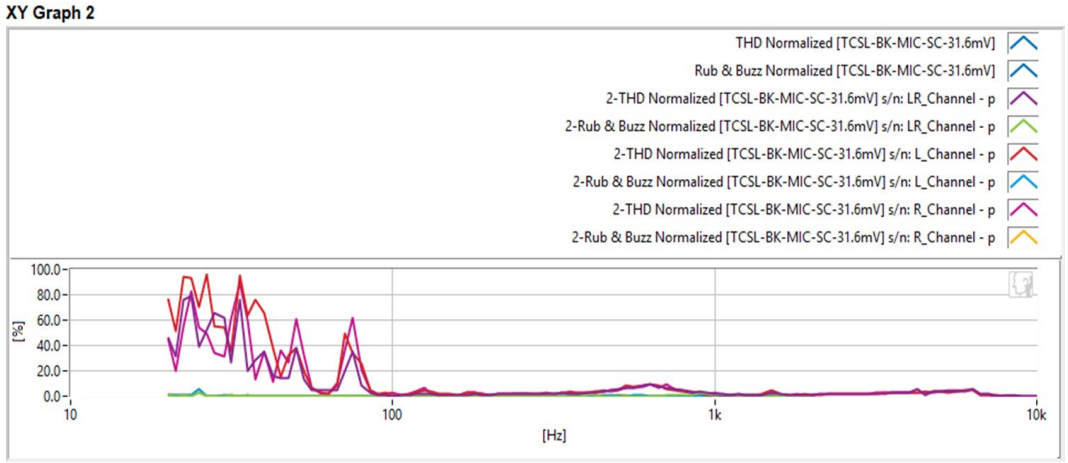
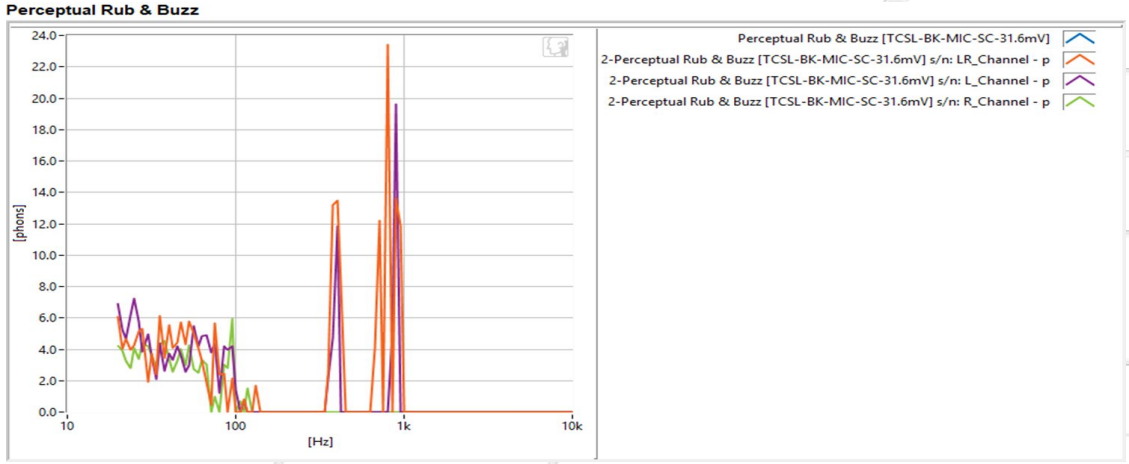


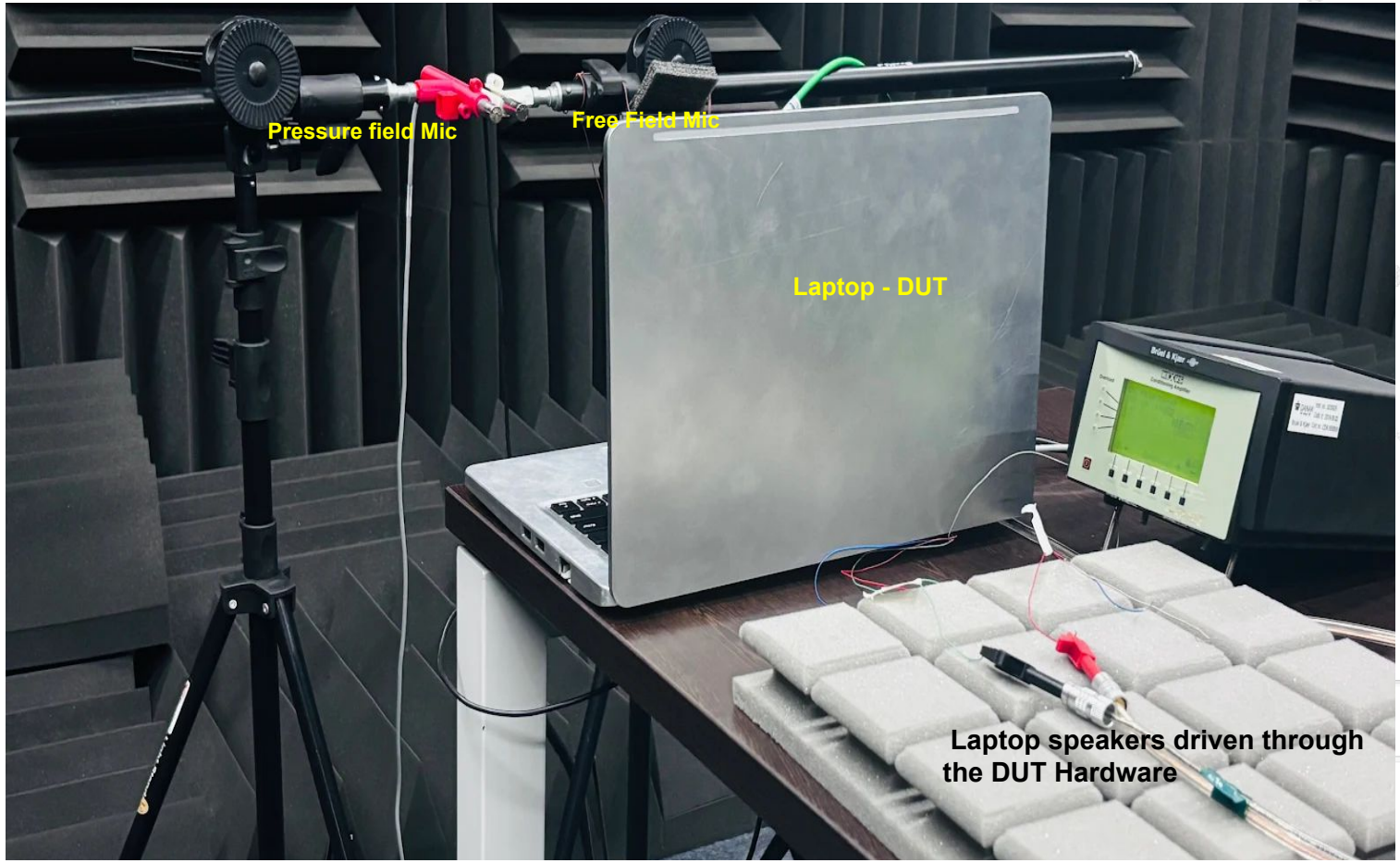
THD Normalized and Rub & Buzz Normalized



Perceptual Rub & Buzz



Test Condition - 0.5m distance,1W Speaker driven by DUT Hardware



Test Type: Direct Speaker Drive Measurement, Rear Radiation / Off-Axis Acoustic Measurement

Distance: 0.5 m between DUT and microphone

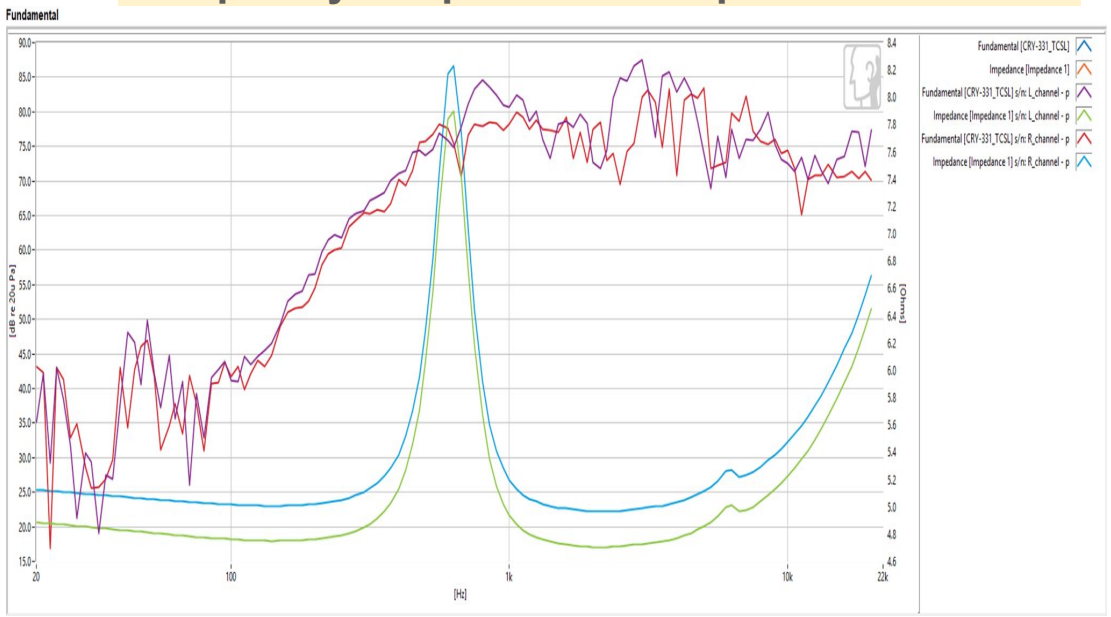
Environment: Acoustic chamber with sound-absorbing wedges

Orientation: Rear side of the laptop chassis facing the microphone

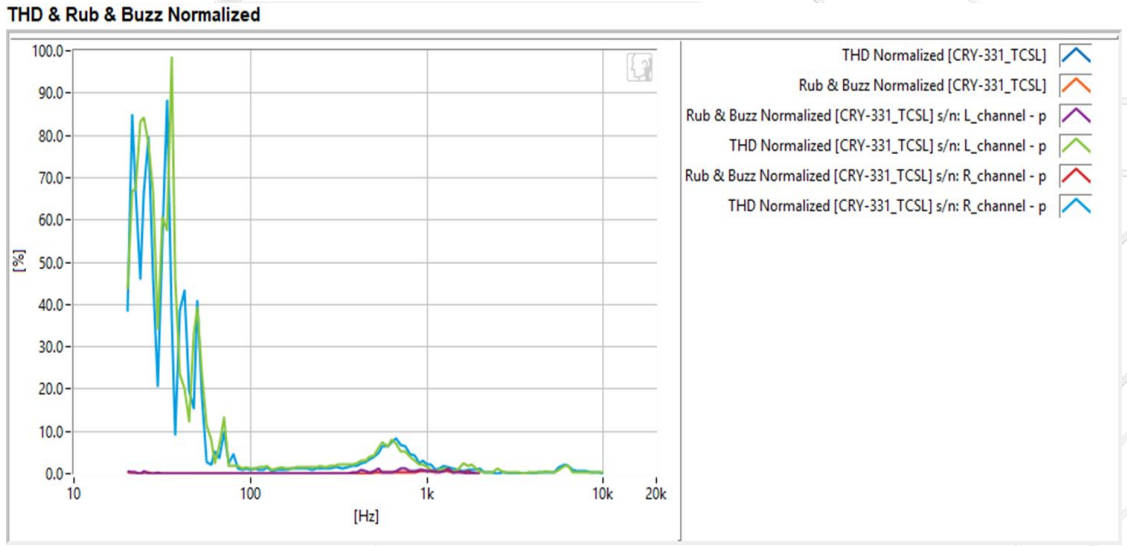
Laptop Condition: Powered off

Acoustic Measurement Results

Frequency Response and Impedance



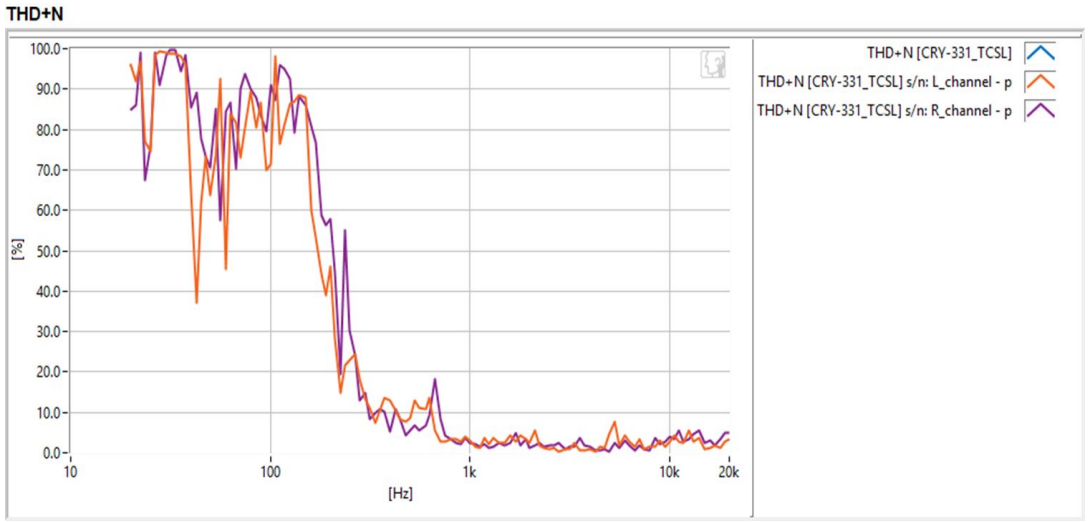
THD - Rub & Buzz



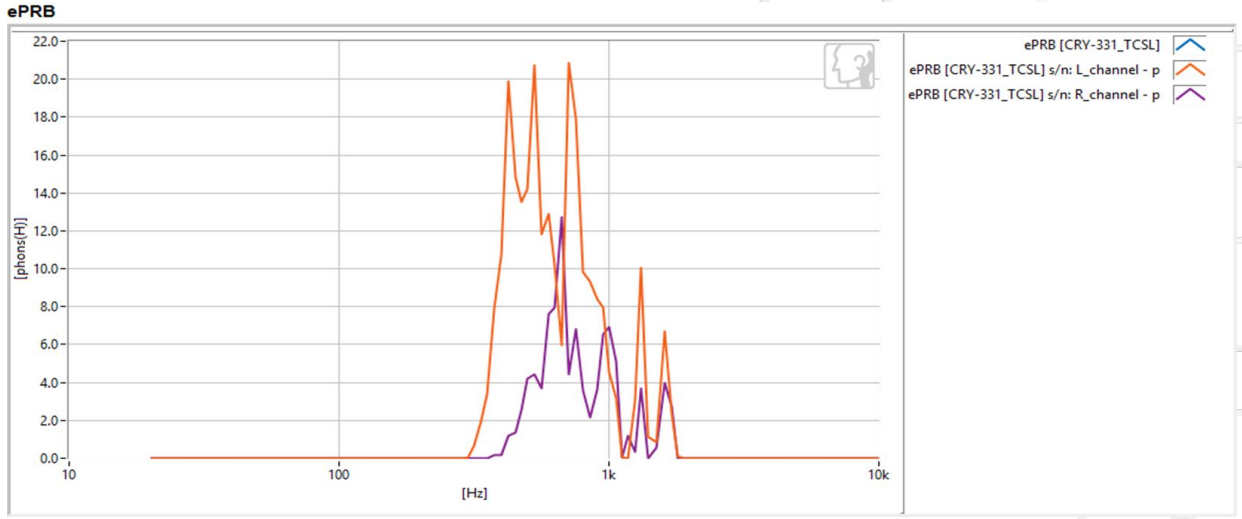
Average Loudness dB SPL

S.NO	Measurement Parameters	Value
1	Average Loudness s/n: Left_Channel	80.7 (dB re 20 uPa)
2	Average Loudness s/n: Right_Channel	77.8 (dB re 20 uPa)
3.	Impedance Est Resonance, L_channel, Y[ohms], Z[Q]	618 Hz, 7.88 Ohms, 2.17
4	Impedance Est Resonance, R_channel, Y[ohms], Z[Q]	619 Hz, 8.21 Ohms, 2.18

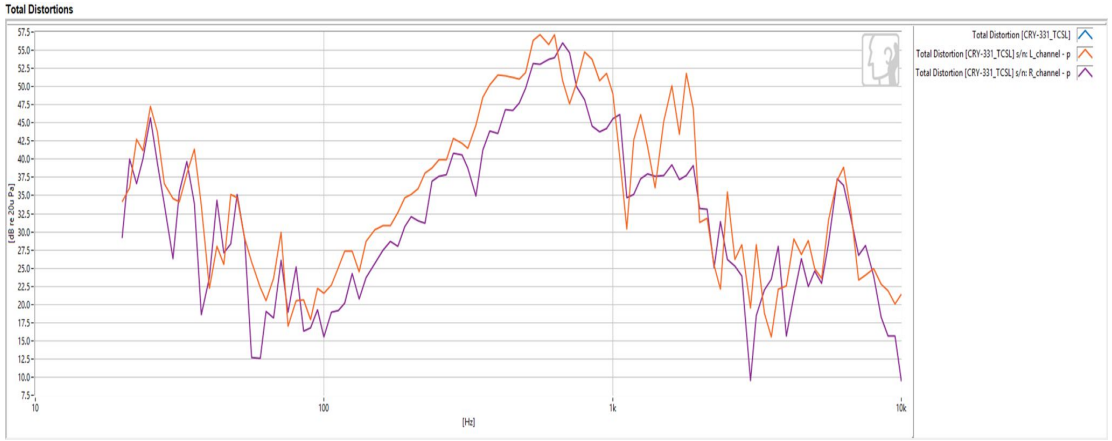
THD+N



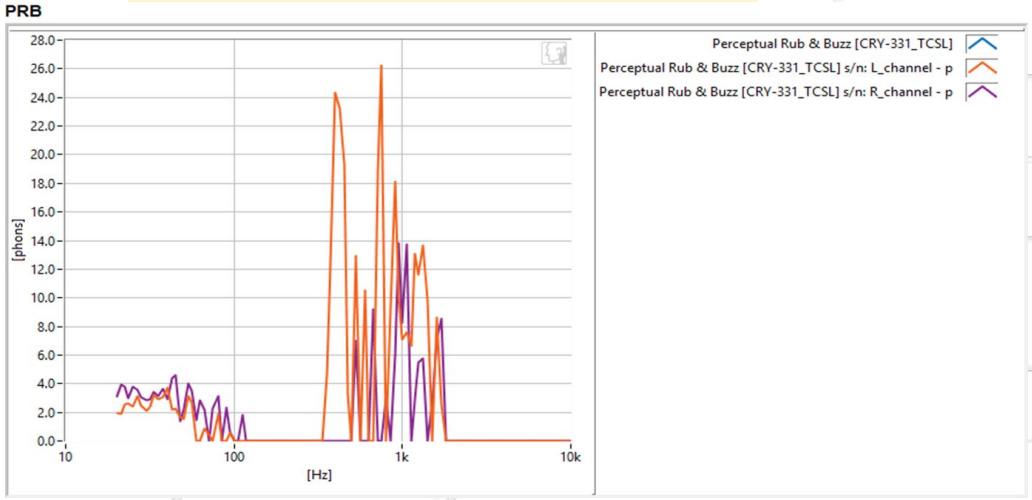
ePRB



Total Distortion



Perceptual Rub & Buzz



Test Condition - 0.5m distance, Speaker driven by DUT Hardware

Test Type: Direct Speaker Drive Measurement,On-Axis Acoustic Measurement

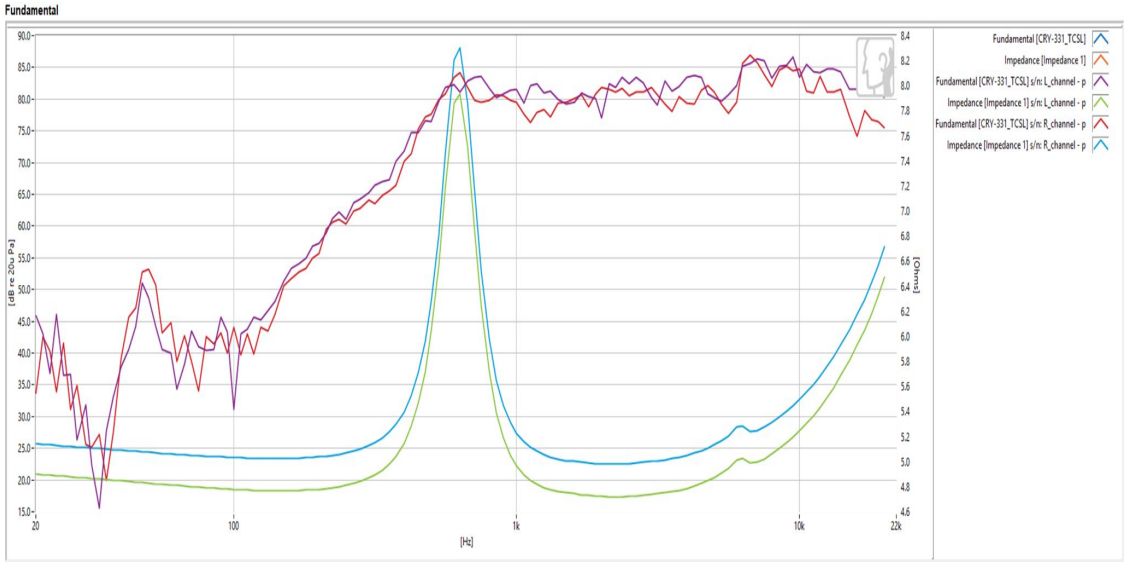
Distance: 0.5 m between DUT and microphone

Environment: Acoustic chamber with sound-absorbing wedges

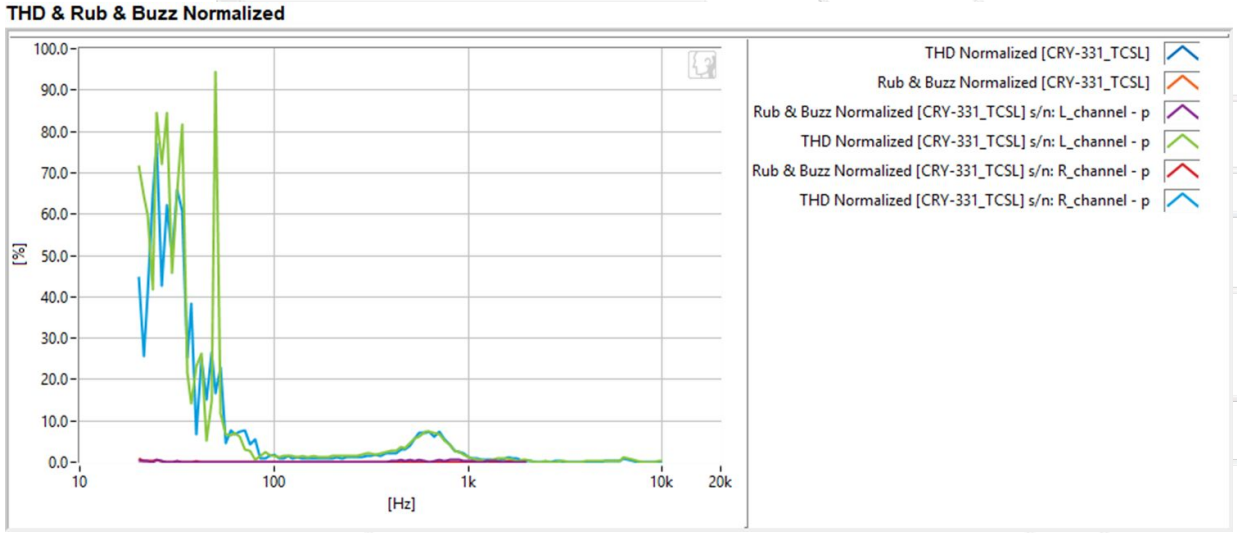
Orientation: Front side of the laptop chassis facing the microphone

Acoustic Measurement Results

Frequency Response and Impedance



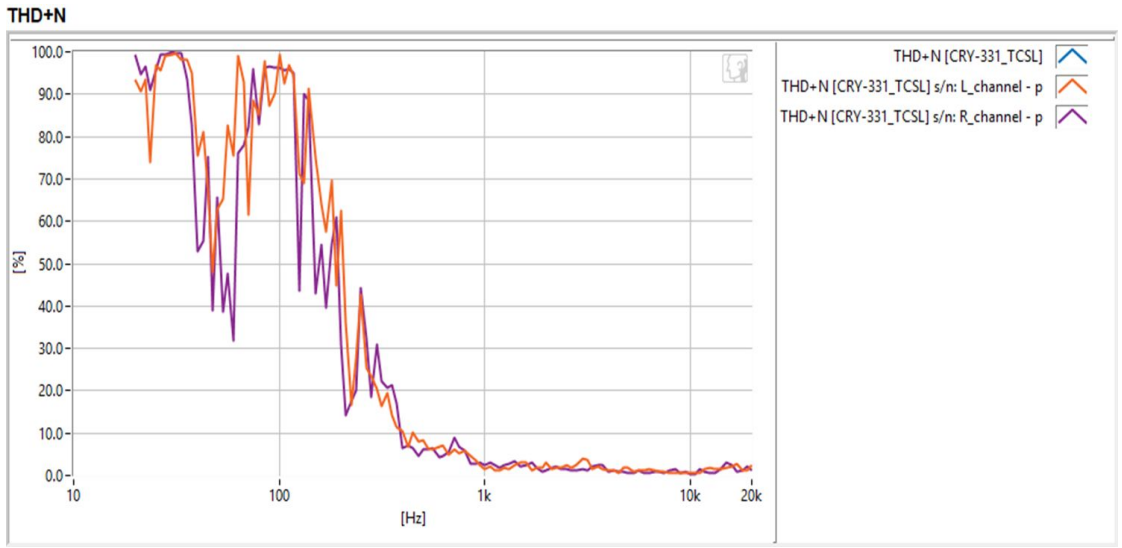
THD - Rub & Buzz Normalized



Average Loudness dB SPL

S.NO	Measurement Parameters	Value
1	Average Loudness s/n: Left_Channel	81.1 (dB re 20 uPa)
2	Average Loudness s/n: Right_Channel	79.2 (dB re 20 uPa)
3.	Impedance Est Resonance, L_channel, Y[ohms], Z[Q]	621 Hz, 7.92 Ohms, 2.16
4	Impedance Est Resonance, R_channel, Y[ohms], Z[Q]	622 Hz, 8.28 Ohms, 2.18

THD+N



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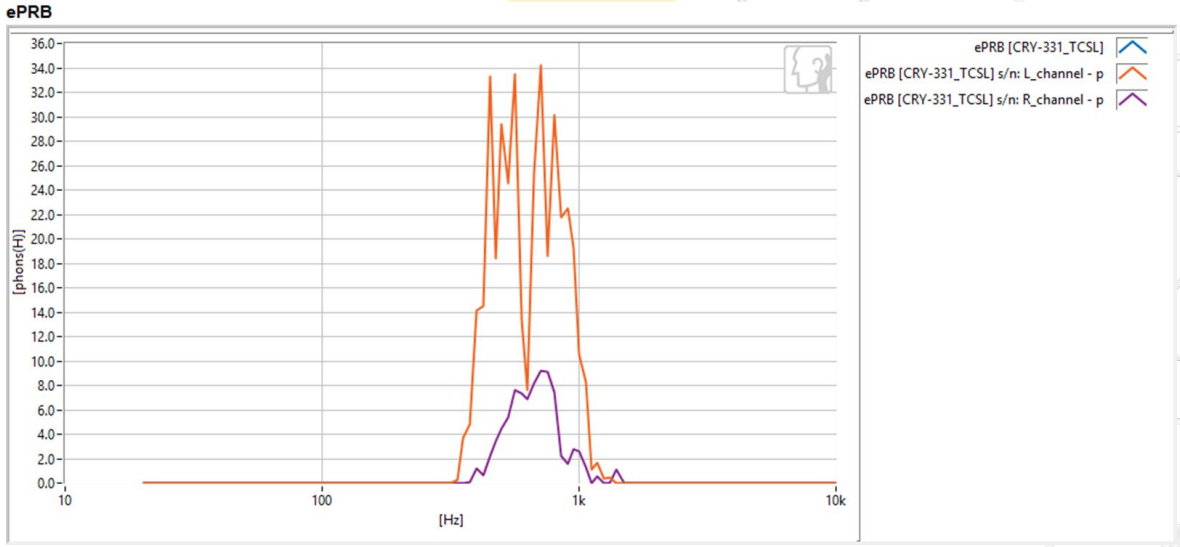
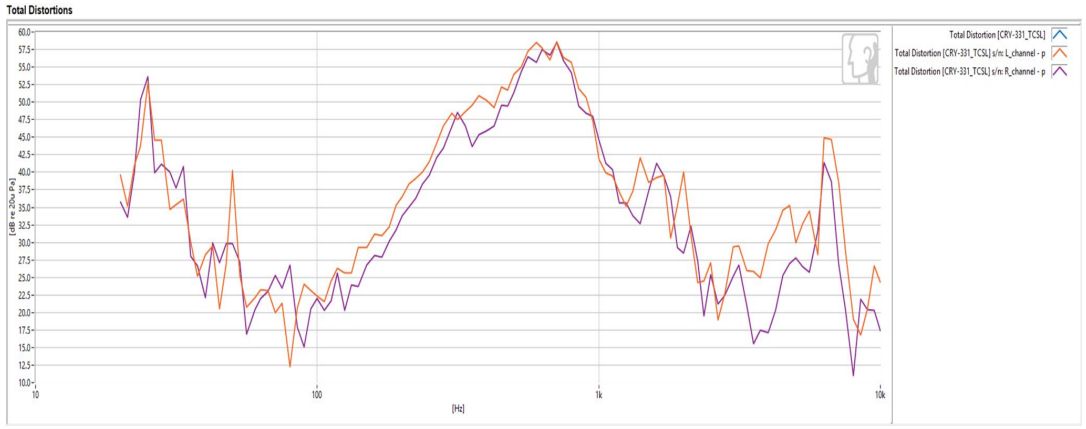
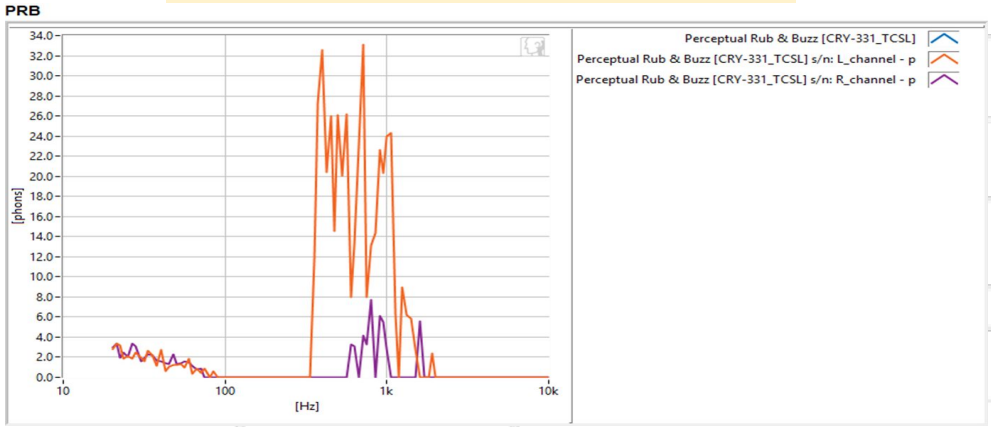


Total Distortion



Perceptual Rub & Buzz



MEASUREMENTS - COMPARISON WITH DATASHEET VALUE

Parameter	Left Channel (Measured) @ 2W, 0.1 m	Right Channel (Measured) @ 2 W, 0.1 m	Expected / Specification 2W, 0.1m
SPL Average of 800, 1000, 1500,2000	87.3	82.736	95 dB
Resonance Frequency (Fr)	613 Hz	615 Hz	630 Hz ± 20%
THD Normalized@ 1 kHz(Disortion)	2.719	5	≤ 5% at 1 W, 0.1 m

Test Observations:

- Laptop speaker acoustic testing conducted at 0.5 m distance in acoustic chamber.
- The right channel SPL was lower than expected due to the longer cable routing inside the laptop.The imbalance has to matched in tuning.
- During initial testing, a right channel polarity issue was identified. After correcting the wire harness as per the schematics, the test results were good.
- For the next batch, the vendor needs to update the wire harness according to the schematics.